

Automatic Plant Watering System

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STATEMENT OF PROBLEM

When my family goes on vacation in the summer holidays, the plants in our garden require water so I decided to make a self watering system for plants that supplies water to the plant by opening a valve when the soil moisture level is low. This way, the plants are watered regularly and there is no dependency on neighbours or others to look after the plants when we are away.

SPECIFICATION

The system should be powered by a 9V battery so that it can be replaced.

The system should pump/pour water onto the soil based on the plant's conditions.

The motor should only turn on if the moisture level is 40% or below and the temperature is 28°C or above.

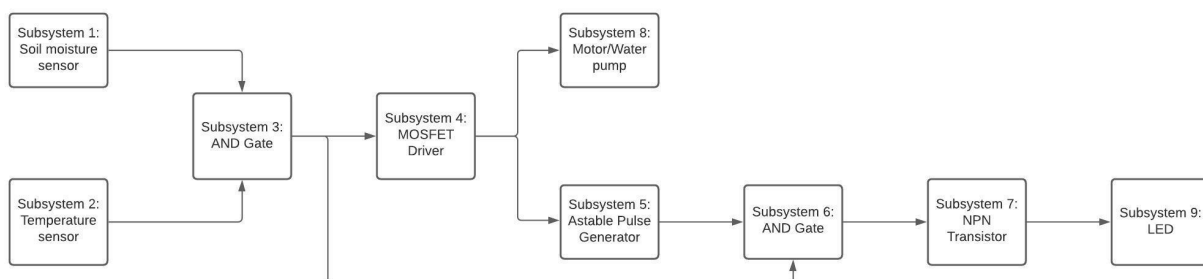
The pulse generator should have a frequency of roughly 1 Hz.

An LED should flash while the motor is running.

The LED should have a brightness of 800 lumens.

The system should have a means of being calibrated to respond to a wide range of conditions.

SYSTEM BLOCK DIAGRAM

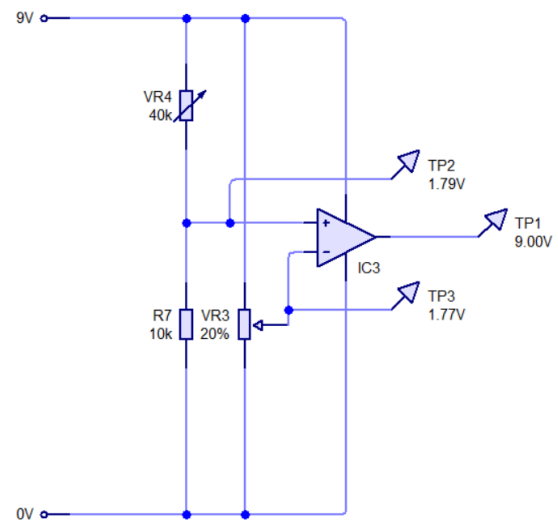
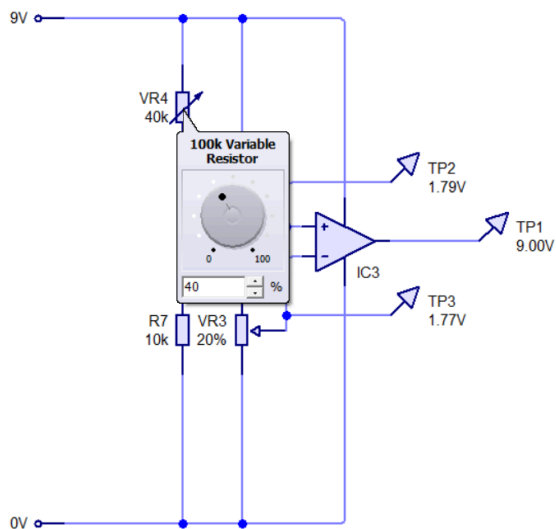


SUBSYSTEM 1 - SOIL MOISTURE SENSOR

Specification

A soil moisture sensor works by measuring how conductive the soil is. By this, I mean that soil with high moisture has a high conductivity but soil with low moisture has a low conductivity. Therefore, the relationship between the soil moisture level and output voltage is directly proportional. Since CircuitWizard doesn't have a soil moisture sensor, I will be using a variable resistor to simulate the expected results (as shown in the below image). I have also added a potentiometer so that the sensitivity of the sensor can be adjusted accordingly where the percentage of resistance is the percentage of moisture. Also, I have calibrated the system such that the potentiometer is kept constant at 20%, with a voltage of 1.77V going into the inverting input of the comparator.

Subsystem circuit diagram



Testing

Test conditions	Moisture percentage	Expected output	Actual output value	Pass/Fail
Input voltage = 1.79V +/- 0.5V	40%	9V +/- 1V	9.00V	Pass
Input voltage = 1.62V +/- 0.5V	45%	0V +/- 1V	-18.24mV	Pass
Input voltage = 1.98V +/- 0.5V	35%	9V +/- 1V	9.02V	Pass

Report on performance

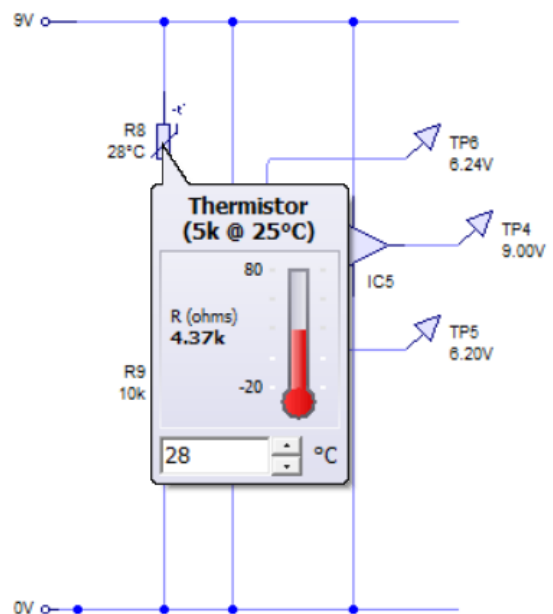
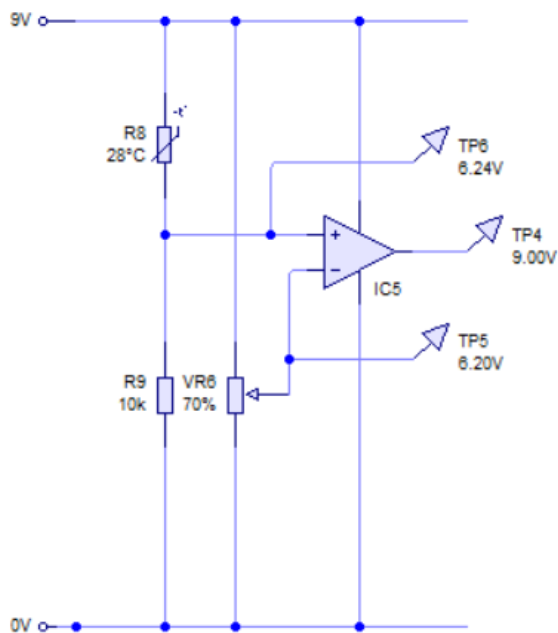
As shown by the results, when a constant voltage of 1.77V enters the inverting input, the comparator compares both inputs and outputs either a high voltage, if the non-inverting input voltage is greater, or a low voltage, if the inverting input voltage is greater.

SUBSYSTEM 2 - TEMPERATURE SENSOR

Specification

The temperature sensor relies on a thermistor that increases resistance as temperature decreases - meaning that the two values are inversely proportional to each other. The thermistor is represented as R8 in the image below. I have also added a potentiometer so that the sensitivity of the sensor can be adjusted accordingly. Also, I have calibrated the system such that the potentiometer is kept constant at 70%, with a voltage of 6.20V going into the inverting input of the comparator.

Subsystem circuit diagram



Testing

Test conditions	Temperature (°C)	Expected output	Actual output value	Pass/Fail
Input voltage = 6.24V +/- 0.5V	28	9V +/- 1V	9.00V	Pass
Input voltage = 5.80V +/- 0.5V	23	0V +/- 1V	-9.19mV	Pass
Input voltage = 6.64V +/- 0.5V	33	9V +/- 1V	9.01V	Pass

Report on performance

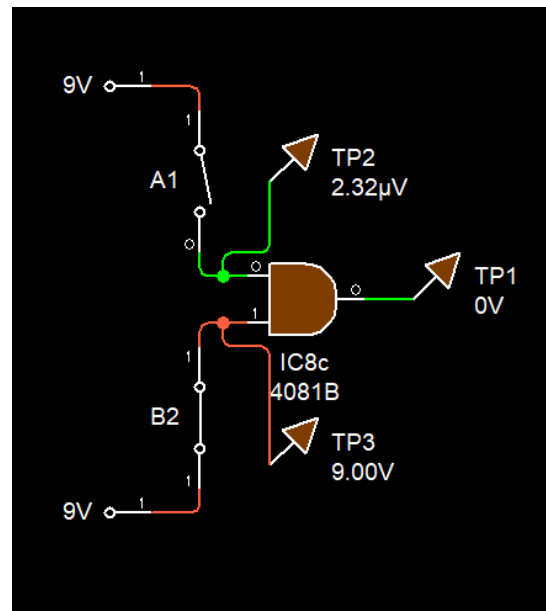
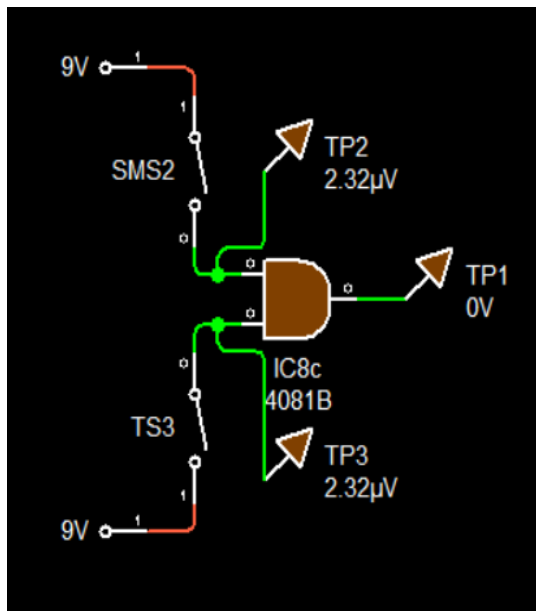
The results show that when temperature is 28°C or above, the output of the comparator is high - showing that the non-inverting input of the comparator was higher than the inverting input (and vice versa).

SUBSYSTEMS 3, 6 - AND GATE

Specification

I included an AND gate in my circuit so that the motor/water pump only runs when both criteria (i.e. low moisture and high temperature) are fulfilled. During my testing, I used 9V power rails and switches to simulate logic levels 0 and 1 coming from my soil moisture and temperature sensors. To note, SMS2 = A1 and refers to the soil moisture sensor and TS3 = B2 and refers to the temperature sensor.

Subsystem circuit diagram



Testing

Test conditions	Input A1	Input B2	Expected output	Actual output value	Output	Pass/Fail
- Input voltage from SMS2 = $2.32\mu\text{V}$ +/- 0.5V - Input voltage from TS3 = $2.32\mu\text{V}$ +/-0.5V	Logic level 0	Logic level 0	0V +/- 1V	0V	Logic level 0	Pass
- Input voltage from SMS2 = 9V +/- 0.5V - Input voltage from TS3 = $2.32\mu\text{V}$ +/-0.5V	Logic level 1	Logic level 0	0V +/- 1V	0V	Logic level 0	Pass
- Input voltage from SMS2 = $2.32\mu\text{V}$ +/- 0.5V - Input voltage from TS3 = 9V +/-0.5V	Logic level 0	Logic level 1	0V +/- 1V	0V	Logic level 0	Pass
- Input voltage from SMS2 = 9V +/- 0.5V - Output voltage from TS3 = 9V +/-0.5V	Logic level 1	Logic level 1	9V +/- 1V	9V	Logic level 1	Pass

Report on performance

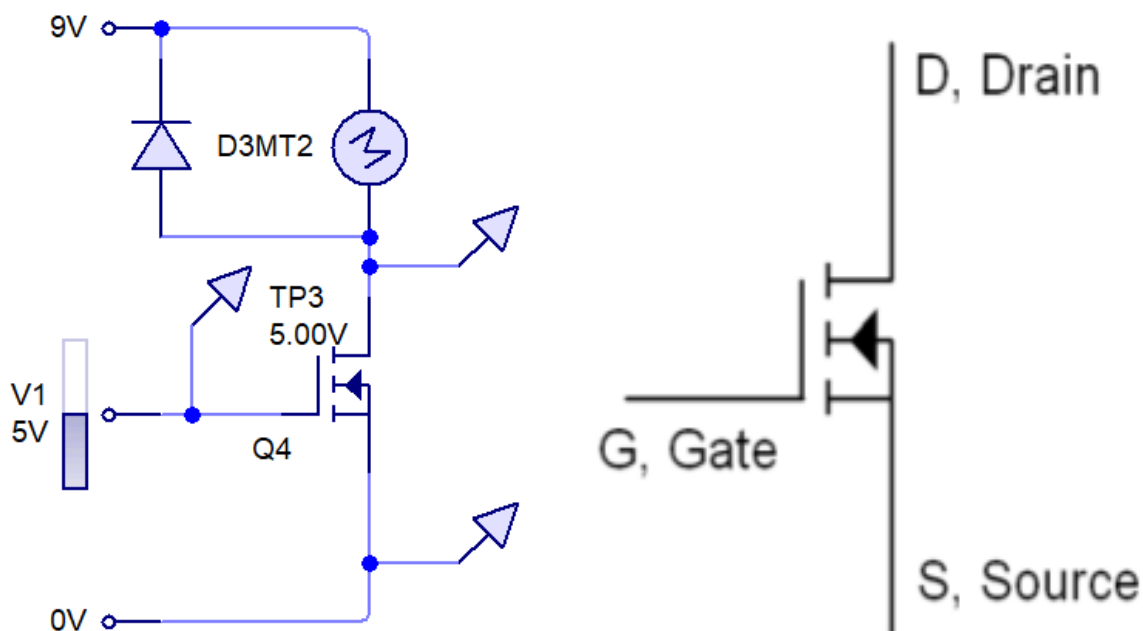
In this test, the voltage value $2.32\mu\text{V}$ indicates a logic level 0 and a voltage value of 9.00V indicates a logic level of 1. Therefore, since AND gates require both inputs to have a logic level of 1, all apart from the last row, have an output of 0V (logic level 0). This is needed for my system because I want both the temperature and soil moisture levels to be sufficient enough for the water pump to start running.

SUBSYSTEM 4 - MOSFET DRIVER

Specification

I have used a MOSFET to amplify both the current and voltage being sent to the water pump. This is so that the motor therefore spins faster because of the higher V_{in} . To test the subsystem I am using a variable input voltage which changes the gate voltage and a motor with test points to measure the voltage. I have also used a protection diode for the motor so that current doesn't flow in the opposite direction. Also, I decided to use a MOSFET instead of an NPN transistor because it is more suitable for high-power applications such as a motor.

Subsystem circuit diagram



Testing

Test conditions	Expected output	Actual output value at drain	Pass/Fail
- Gate voltage = 0V +/- 0.5V	9V +/- 1V	9.00V	Pass
- Gate voltage = 2.5V +/- 0.5V	8V +/- 1V	8.80V	Pass
- Gate voltage = 5V +/- 0.5V	6V +/- 1V	5.76V	Pass
- Gate voltage = 9V +/- 0.5V	3V +/- 1V	3.21V	Pass

Report on performance

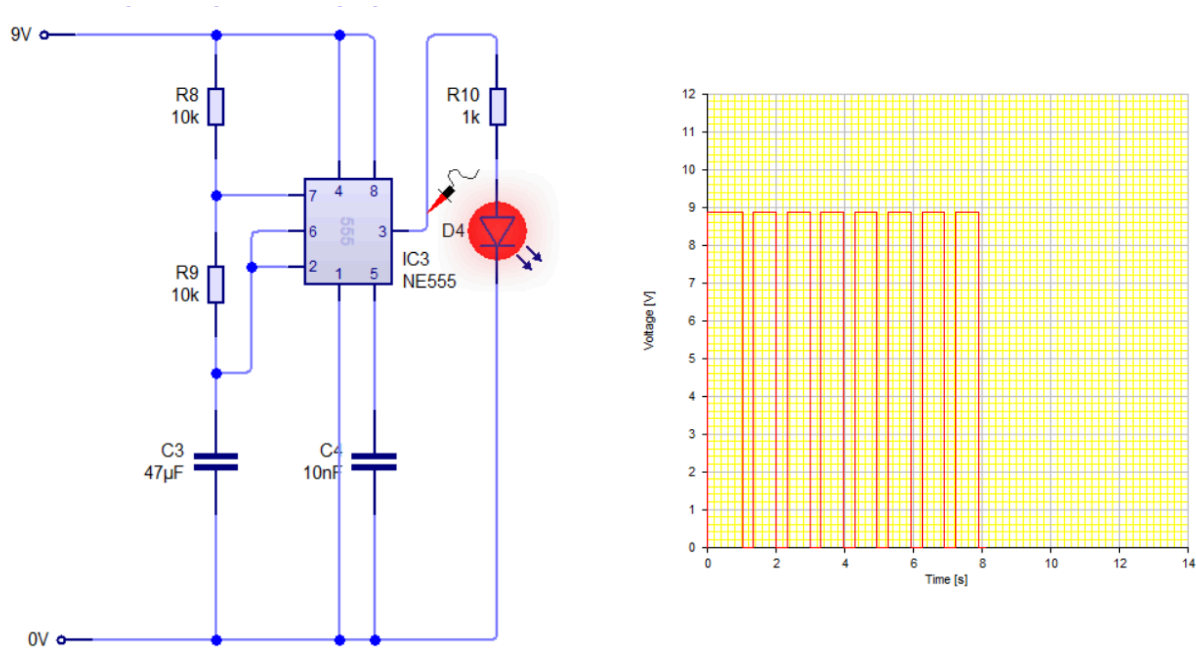
When I conducted the test, I found that the voltage was amplified and the g_m value was around 0.12 siemens +/- 1S. This is due to the equation, " $g_m = I_D/V_{GS}-3$ ". Because of this equation, the minimum voltage for the MOSFET to rotate the motor is 3 volts or above as the denominator will be negative, otherwise. Also, I expected a value of 9V for all outputs except for the one where gate voltage was 0V because I was using a 9V power supply which causes voltage to flow down to the drain pin.

SUBSYSTEM 5 - ASTABLE PULSE GENERATOR

Specification

For the duration that my water pump is on, I want an LED to flash. I did this by connecting the pins of the monostable such that the output voltage switches between logic level 1 and 0. Also, I want the astable to have an output frequency of about 1 Hz where the time the output is high is 651.420 milliseconds and the time the output is low is 325.710 milliseconds. Therefore, the mark to space ratio is 2:1. To simulate the output, I included an LED as well (along with a limiting resistor to protect the LED from large currents).

Subsystem circuit diagram

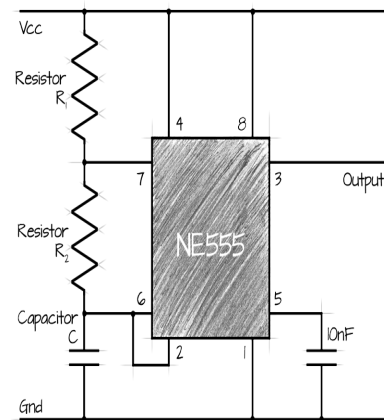


Testing

Test conditions	Expected output	Actual output value	Pass/Fail
$R_1 = 10k\Omega$ $R_2 = 10k\Omega$ $C = 47\mu F$	1 cycle lasting for 1 second +/- 1 second	1 cycle lasting for 1.023 seconds	Pass

Report on performance

As shown by the results, the actual time that one whole cycle took was 1.023 seconds. I calculated this frequency by converting the units correctly and using the formula: $f = 1.44 / C(R_1 + 2R_2)$. The values of C , R_1 and R_2 in my circuit can be correlated with the template on the right. Also, the graph shows a voltage of 8.86V when high and a voltage of 0V when low. Lastly, the graph shows a longer time on high for the first cycle but this is only because the circuit is adjusting and as seen above, the timings are corrected immediately for the upcoming cycles.

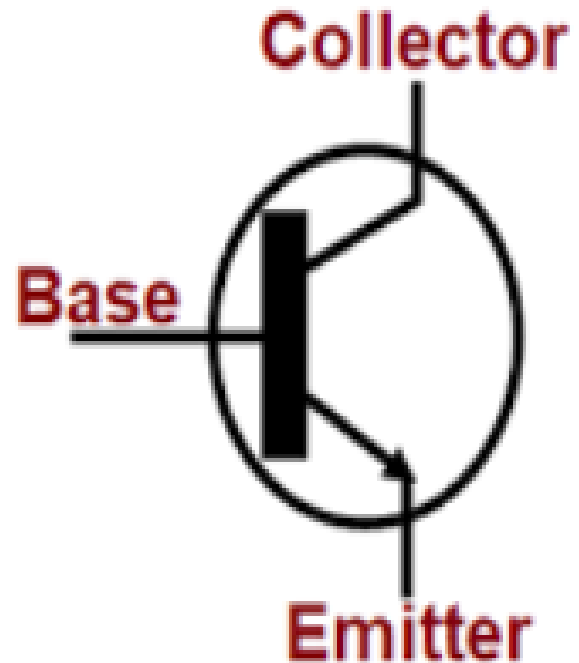
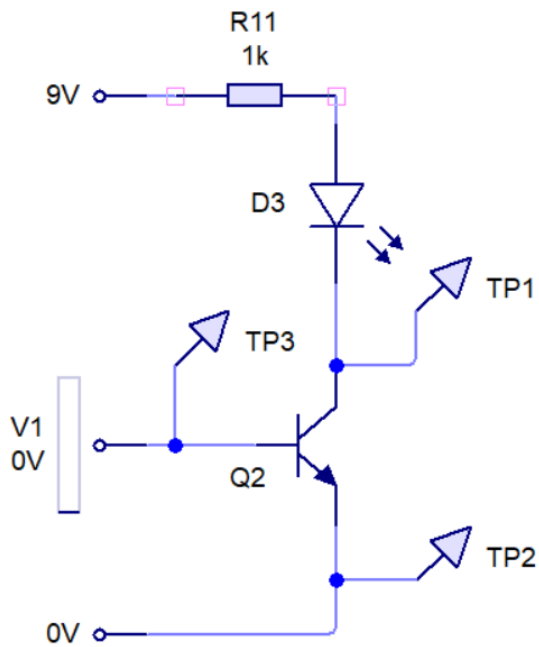


SUBSYSTEM 7 - NPN TRANSISTOR

Specification

I have used a transistor to amplify the voltage because the LED that receives the amplified voltage is a low power application. Another reason I used a transistor is because the brightness of the LED was too low without one. To test the subsystem I am using a variable input voltage which changes the voltage entering the base pin of the transistor and an LED with test points to measure the voltage. I have also used a limiting resistor for the LED to prevent too much current from flowing into the LED.

Subsystem circuit diagram

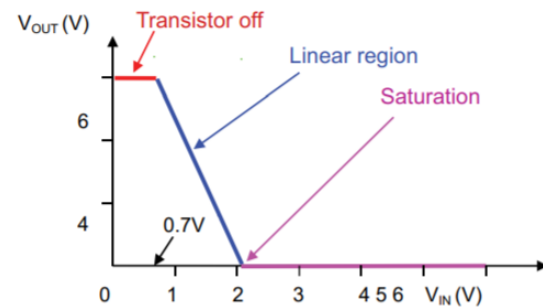


Testing

Test conditions	Expected output	Actual output value at collector	Pass/Fail
- Base = 0V +/- 0.5V, 0A	9V +/- 1V, LED off	8.50V, 0A, LED off	Pass
- Base = 2V +/- 0.5V, 929.44mA	0.5V +/- 1V, LED on	576.29mV, 6.52mA, LED on	Pass
- Base = 4V +/- 0.5V, 2.62A	2V +/- 1V, LED on	1.60V, 5.65mA, LED on	Pass
- Base = 6V +/- 0.5V, 4.33A	3V +/- 1V, LED on	2.63V, 4.77mA, LED on	Pass

Report on performance

The results show that when base voltage is 0V, the output voltage at the collector pin is 8.5V. This could be because I added a limiting resistor from ground which is preventing an output voltage of 9V. Also, the LED is off and this is because, for NPN transistors, the saturation point is when V_{in} is 0.7V (as shown on the right). For the remainder of the results, they are as expected. Lastly, I found that the h_{FE} (current gain) of the transistor was around 0.003. I calculated this by using the equation: $h_{FE} = I_C/I_B$ and then by finding the average from the three tests I conducted above.

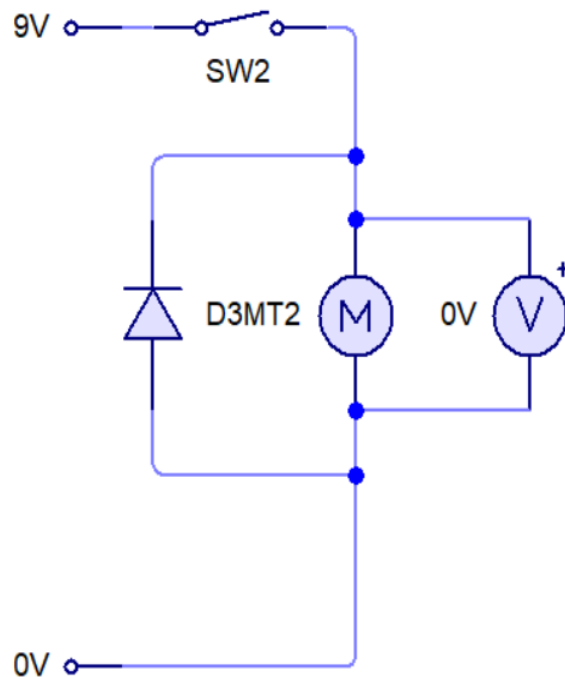


SUBSYSTEM 8 - MOTOR

Specification

Since Circuit Wizard doesn't have a water pump that I intend to use, I will be simulating with a motor - which is the closest available component to a water pump. I have also used a protection diode for the motor so that current doesn't flow in the opposite direction. The simulation will consist of the motor connected to a power source, a protection diode, a switch and a voltmeter to measure the voltage across the motor. To note, the switch simulates the output of the MOSFET when the soil moisture sensor and temperature sensors detect a change.

Subsystem circuit diagram



Testing

Test conditions	Expected output	Actual output value	Pass/Fail
- Switch off, 0V	0V +/- 1V	0V	Pass
- Switch on, 9V	9V +/- 1V	8.90V	Pass

Report on performance

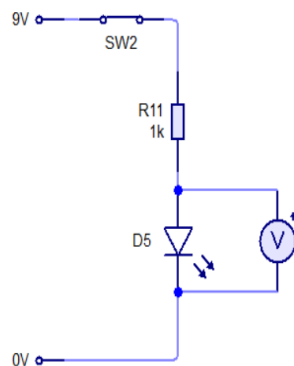
The results show that when the switch is closed and the input voltage is 9V, the motor runs and when the switch is open and the input voltage is 0V, the motor doesn't run. To note, the switch simulates the output of the MOSFET when the soil moisture sensor and temperature sensors detect a change.

SUBSYSTEM 9 - LED

Specification

I am simulating the LED once the astable has outputted a voltage. To simulate a low output voltage from pin 3 of the astable, I am using an open switch and to simulate a high output voltage from pin 3, I am using a closed switch. Also, I have added a limiting resistor to prevent high currents from flowing through the LED. The simulation will consist of an LED connected to a power source, a limiting resistor, a switch and a voltmeter to measure the voltage across the LED.

Subsystem circuit diagram



Testing

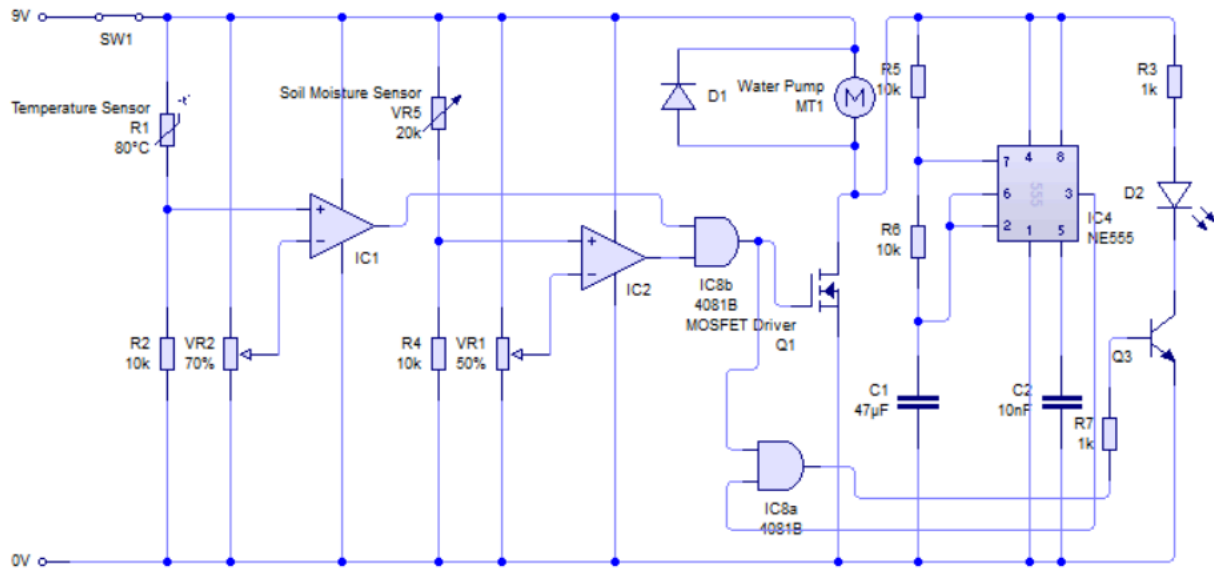
Test conditions	Expected output	Actual output value	Pass/Fail
- Switch off, 0V	0V +/- 1V, LED off	0V	Pass
- Switch on, 9V	1V +/- 1V, LED on	1.98V	Pass

Report on performance

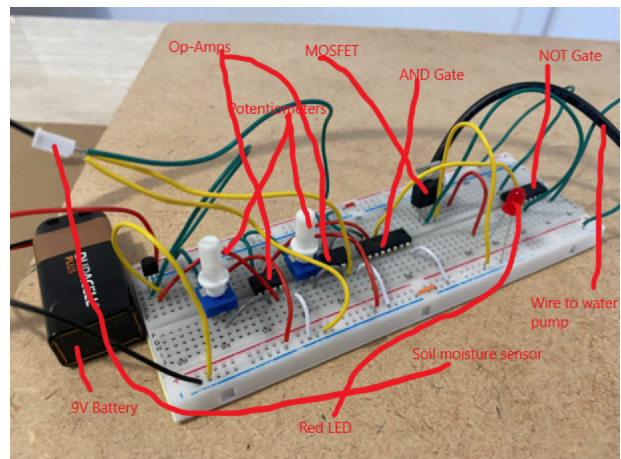
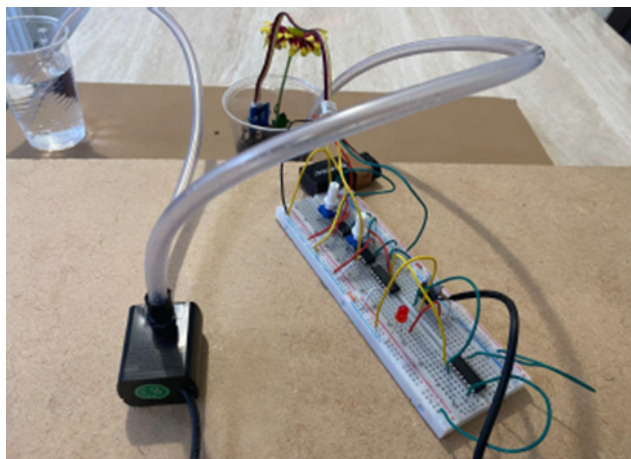
The results show that when the switch is closed and the input voltage is 9V, the LED is on and when the switch is open and the input voltage is 0V, the LED is off. To note, the switch simulates the output of the astable when high and when low.

COMPLETED CIRCUIT

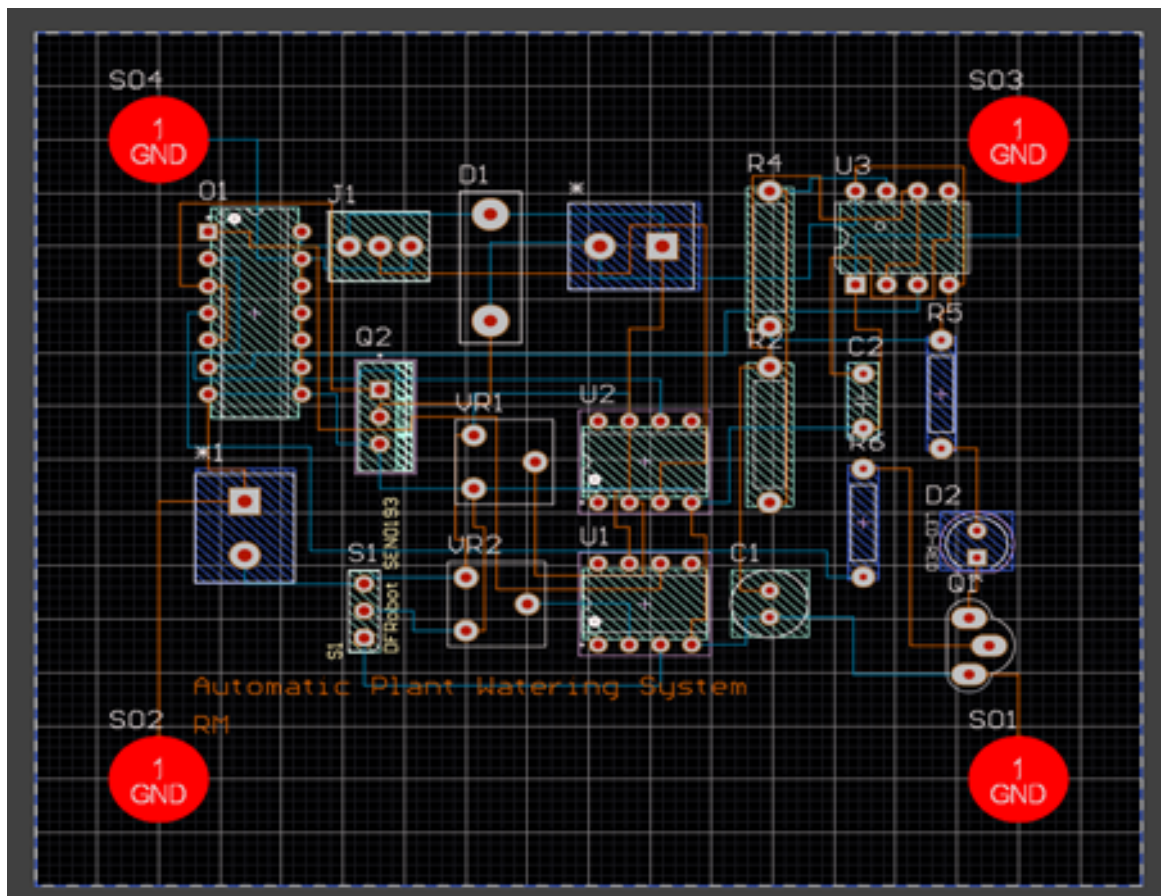
Circuit Diagram



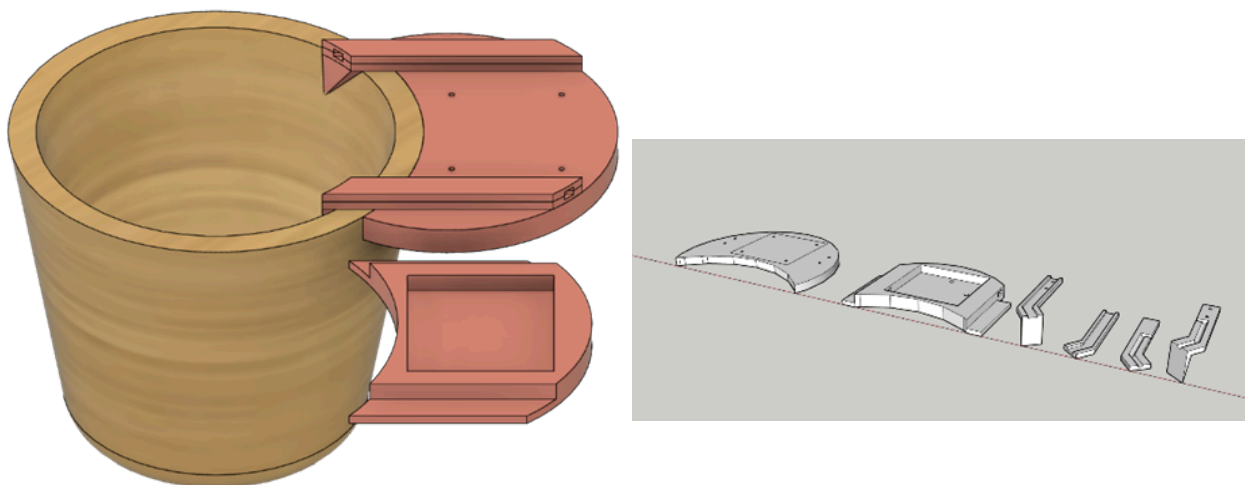
Breadboard Prototype



PCB Design on Altium CircuitMaker



Housing Design on Fusion 360



COMPONENT LIST

Component Name	Quantity
Op-Amp	2
Temperature Sensor	1
Soil Moisture Sensor	1
Potentiometer	2
AND Gate	2
N-Channel MOSFET	1
Water Pump	1
Diode	1
NE555 Timer	1
NPN Transistor	1
LED	1
10k Resistor	2
1k Resistor	1
47 μ F Capacitor	1
10nF Capacitor	1

TESTING OF THE ENTIRE SYSTEM

Temperature	Moisture percentage	Expected output	Actual output value	Pass/Fail
20°C	35%	Motor and LED are off	Motor and LED are off	Pass
20°C	40%	Motor and LED are off	Motor and LED are off	Pass
20°C	45%	Motor and LED are off	Motor and LED are off	Pass
28°C	35%	Motor and LED are off	Motor and LED are on	Pass
28°C	40%	Motor and LED are off	Motor and LED are on	Pass
28°C	45%	Motor and LED are off	Motor and LED are off	Pass
30°C	35%	Motor and LED are off	Motor and LED are on	Pass
30°C	40%	Motor and LED are off	Motor and LED are on	Pass
30°C	45%	Motor and LED are off	Motor and LED are off	Pass

USER GUIDE

First, a 9V battery must be connected to the circuit. Once both the soil moisture sensor probes and the temperature sensor are in the soil, the calibration sequence can be initiated. The soil moisture sensor should be calibrated by rotating the potentiometer to the point when the water pump starts to run. This calibrates the sensor such that when moisture is below the calibrated level, the water pump will turn on. Next, the temperature sensor should be calibrated by rotating the potentiometer to the point where the water pump starts to run. This process makes sure that the water pump turns on if the temperature is above the calibrated level. Once that is done, the PCB can be mounted onto the circuitry housing and the vacuum end of the water pump's pipe can be inserted into a water source (e.g. a water bottle). Lastly, once at least one cycle has run, the system is ready to be used.

EVALUATION

Evaluation of how the system works

The system works by detecting a change in the temperature or soil moisture levels. More specifically, when temperature is above or equal to 28°C and the soil moisture level is below or equal to 40%, the water pump will allow water to flow into the soil and hence raise the soil moisture level and lower temperature. This is done because a HIGH output is outputted by the temperature sensor when temperature is above or equal to 28°C and a LOW output is outputted by the soil moisture sensor when soil moisture levels are below or equal to 40% so an AND gate then only outputs HIGH when both of these inputs are HIGH. The HIGH output of the AND gate then goes through a MOSFET to a water pump and pulse generator. The configuration of the astable's pins causes the astable to be triggered when the input is HIGH. Therefore, when a HIGH input enters the astable, this causes the LED to flash while the soil moisture level is low and temperature is high. Then, once the levels are optimal, the LED stops flashing and the water pump stops running as both sensors would output LOW.

Evaluation of completed system

The system works as expected and I think that to improve the system, I could potentially use an OR gate instead of an AND gate because it might just be that the plant is located in an environment where temperature is high. Also, instead of a temperature sensor, I could use a humidity sensor placed onto the underside of a leaf of the plant so that the sensor can monitor when the water released by the plant is low and hence water the plant to increase transpiration rates. Also, there was no way of testing the brightness of the LED on Circuit Wizard so I resulted to detecting the brightness by eye which is not very accurate.